Project Design Document

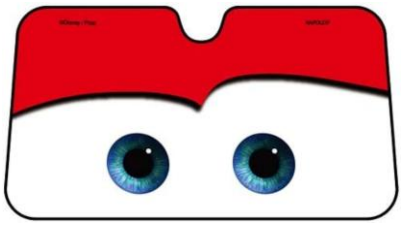
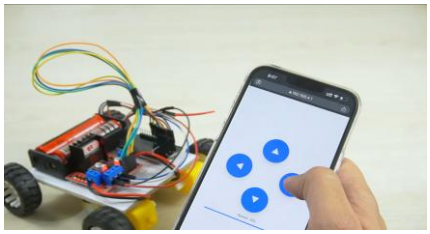
Design Intent

Experience:
Create a playful, responsive toy car that goes beyond basic control by using light and sound to enhance interaction. The synchronized LED indicators and adaptive buzzer feedback aim to create an engaging and intuitive experience, while also building a sense of nostalgia inspired by character-based vehicles. The addition of an automatic stop during close-range reversing improves safety and realism.

Target Users:
Kids and fans of Cars, or anyone who enjoys interactive, app-controlled toys.

Inspiration & References

https://youtu.be/Tta4xhmYSwI?si=WCV5FQTvNgDemQX9

- Inspired by Cars for character-like vehicles and visual identity
- Existing RC cars and DIY ESP32 robot cars for movement and control systems

Learnings:

- We learned how to use a mobile app as a remote control instead of building a physical controller
- We understood how ESP32 can be used to build interactive connected toys
- We explored how light and sound can communicate system feedback (turning, reversing, proximity) in an intuitive way

User Journey

1. User opens the mobile app and connects to the car via WiFi
2. User presses directional buttons to move the car (forward, backward, left, right)
3. System responds instantly with motion using 4 DC motors controlled by 2 motor drivers
4. When the user presses left or right, the car turns and LED indicators automatically blink in sync with the direction
5. When the user presses reverse, the car moves backward and:

- The buzzer starts emitting a reverse warning sound
- LEDs blink at 0.5-second intervals

6. As the car approaches an obstacle (using rear ultrasonic sensor):

- > 15 cm: Slow beeping pattern
- 10–15 cm: Medium speed beeping
- 5–10 cm: Faster beeping
- < 5 cm: Continuous beep

7. Safety Feature:

- When distance is below 5 cm, the car automatically stops
- Backward motion is disabled to prevent collision

8. The system continuously provides feedback through motion, light, and sound, making interaction intuitive, safe, and engaging

Success Criteria

Core Questions:

1. Does the car move correctly when a button is pressed?
2. Do the LEDs and buzzer respond at the right time?
3. Is the system easy and intuitive to use?

Functional

1. Car responds to movement commands with minimal delay
2. LED indicators synchronize correctly with left/right turning
3. LEDs blink at consistent 0.5-second intervals during reverse
4. Buzzer activates during reverse and changes frequency based on distance
5. Ultrasonic sensor reliably detects obstacles in real time
6. **Car automatically stops when object is closer than 5 cm during reverse**

User Experience

1. Controls are simple and intuitive through the mobile app
2. Clear and immediate feedback through LEDs and buzzer
3. Reverse assist system improves awareness and usability
4. Safety stop feature increases reliability and user confidence
5. System is durable, stable, and safe for repeated use

System Overview

The system is a WiFi-controlled smart toy car built using an ESP32 as the central controller. A mobile app sends commands over WiFi, which the ESP32 processes to control different components. Four DC motors, managed through two motor drivers, handle movement.

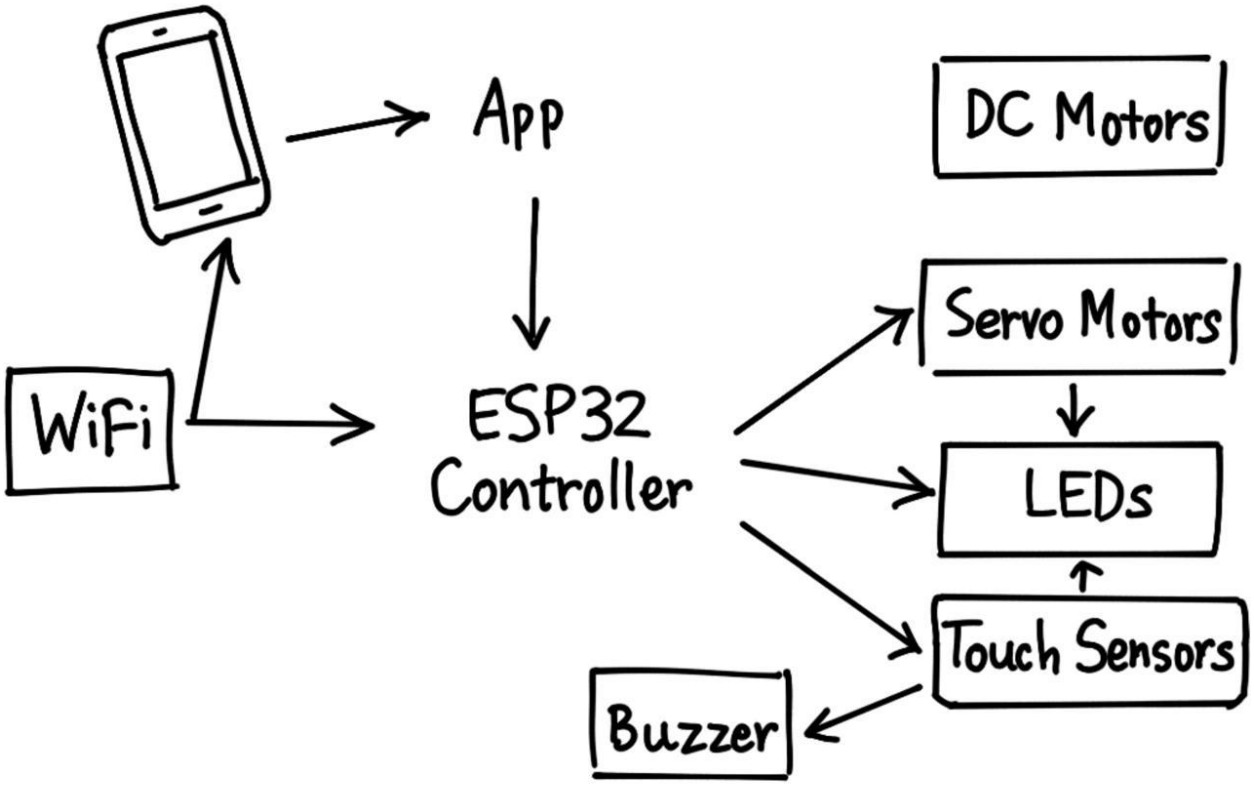
LED lights function as indicators and are synchronized with turning and reversing actions. During reverse, LEDs blink while a buzzer provides adaptive feedback based on distance from obstacles detected using an ultrasonic sensor placed at the rear.

A key feature is the **automatic safety stop**, where the car halts when an object is detected within 5 cm, preventing collision during reverse motion.

The physical structure of the car is made using a **cardboard body and base**, allowing easy prototyping, lightweight construction, and flexible integration of components.

The system integrates motion, light, and sound to create an interactive and responsive toy experience inspired by Cars.

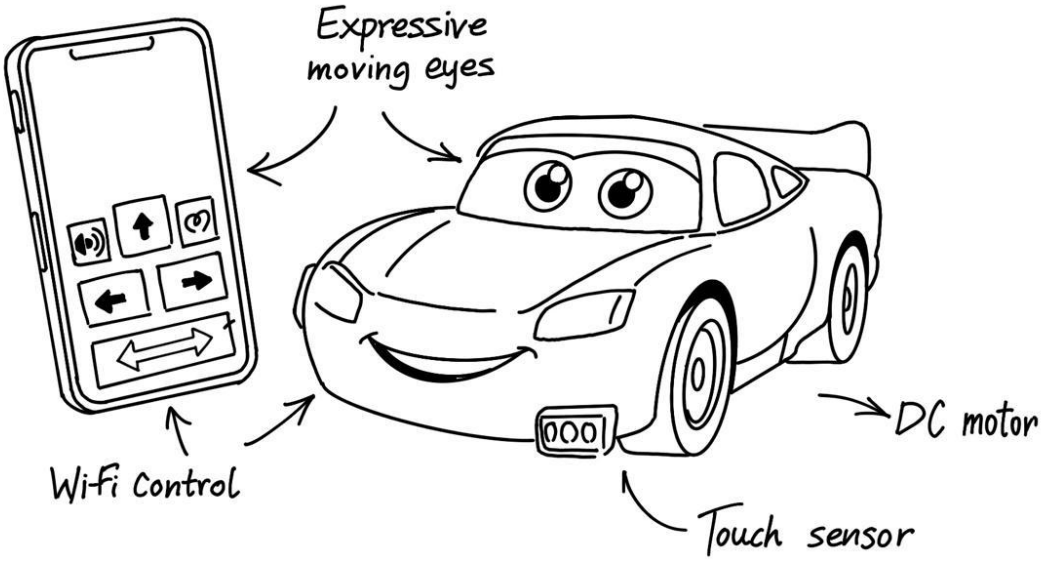
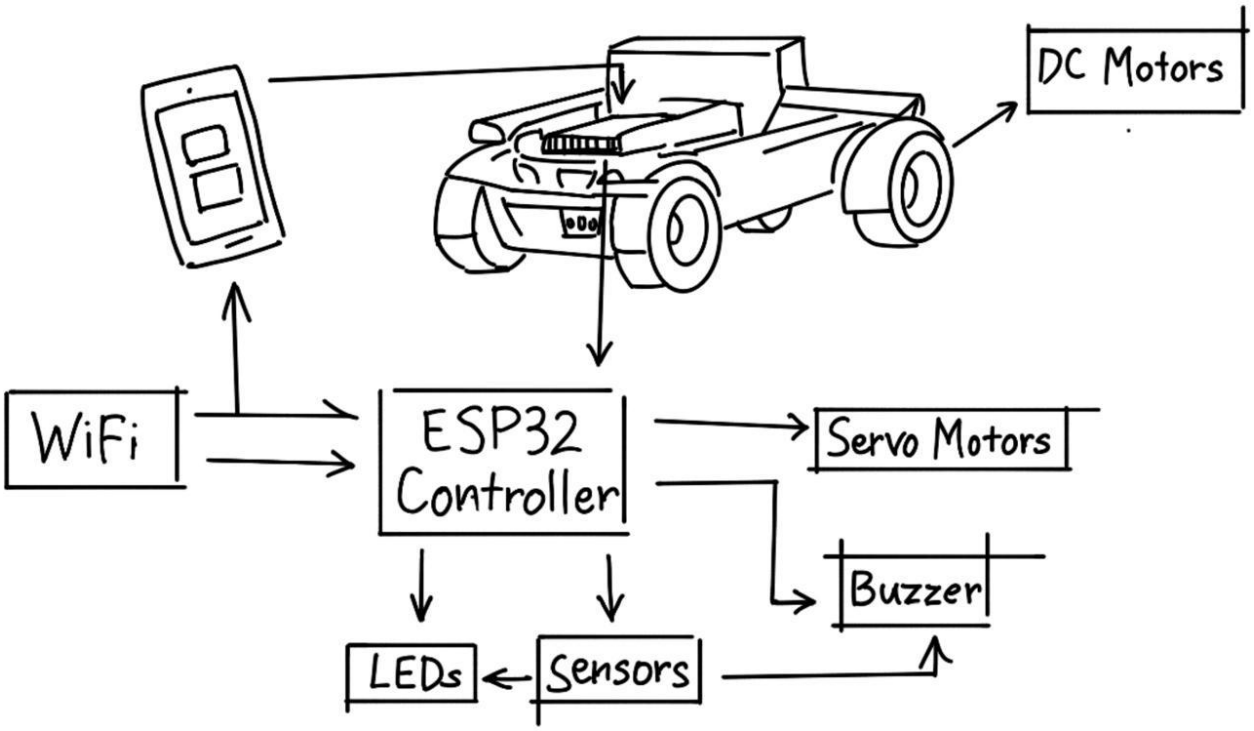
High Level Block Diagram



Input-Output Mapping

Input	Process	Output
App button (F/B/L/R/Stop)	Signal mapped to motor driver pins	Car moves accordingly
Direction (Left/Right)	LED control logic triggered	Indicators blink
Reverse command	Activate sensor + timing logic	LEDs blink + buzzer activates
Ultrasonic sensor	Distance calculated	Buzzer frequency changes
Distance < 5 cm	Safety condition triggered	Car stops + continuous beep

Sketches and Concepts



Final System Architecture